



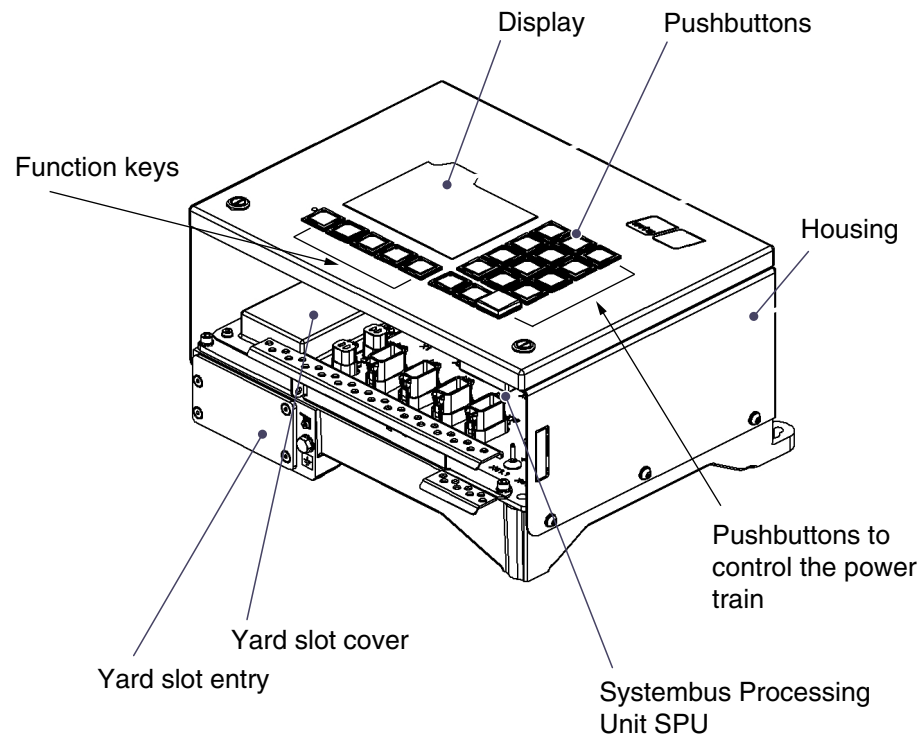
Please read and observe the detailed overall MTU documentation for BlueVision_Advanced | NewGeneration before using this Quick Reference Guide! Pay particular attention to the safety notes in the overall documentation.

Local Operating Panel LOP 14

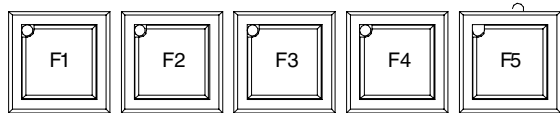
The Local Operating Panel is the monitoring and control unit for the engine and gearbox*.

An alphanumeric color screen and a combined control and indicator panel are provided to facilitate operation and indication.

Measured values and alarm messages in case of limit violations are displayed in accordance with MTU specifications.



Function keys

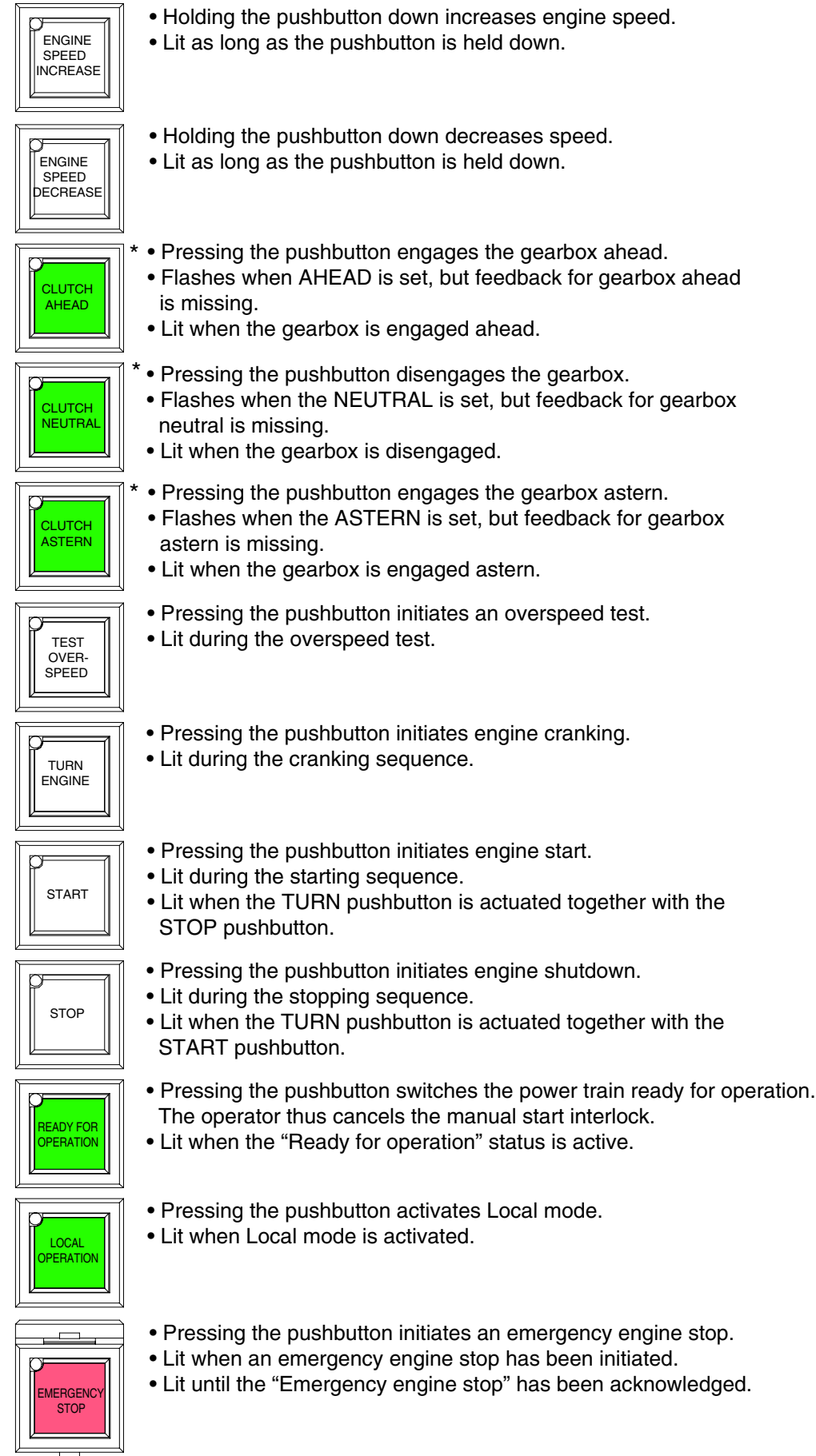


Function keys F1 to F5 are used to control and operate the graphic screen pages on the display, to acknowledge alarms and for dimming.

The various functions of the keys are indicated by graphics and the available functions depend on which page is currently on display.

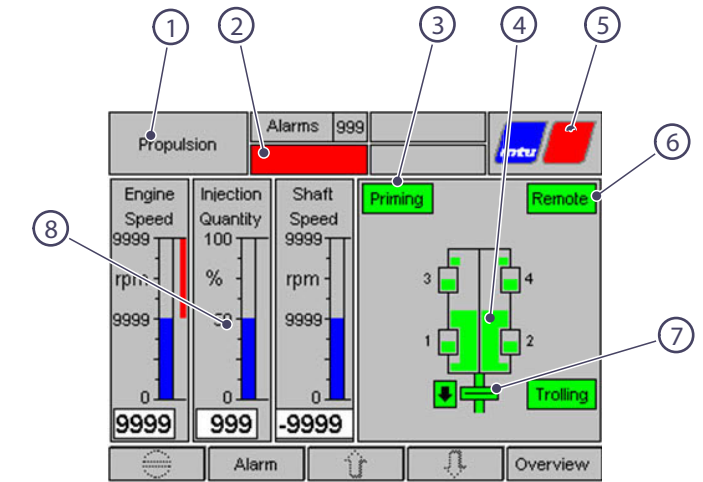
Each key is lit as long as it is held down.

Pushbuttons to control the power train



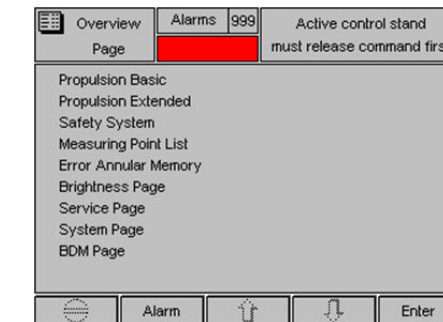
* Optional/depending on plant configuration

Screen pages on Local Operating Panel LOP 14

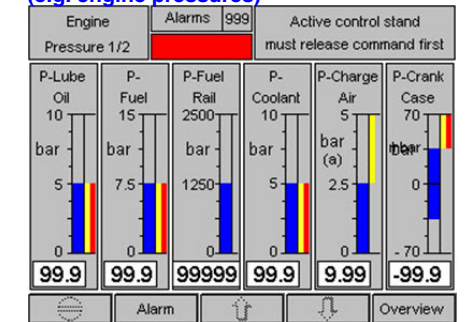


- | | |
|---|------------------------------------|
| 1 Screen page name | 5 Message window |
| 2 Active alarm indicator | 6 Local/Remote control active |
| 3 Oil priming active | 7 Gearbox status/Trolling active |
| 4 Engine and exhaust turbocharger indicator | 8 Measuring point barchart display |

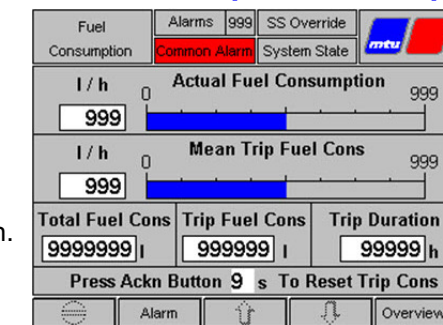
Overview page



Engine screen page (e.g. engine pressures)



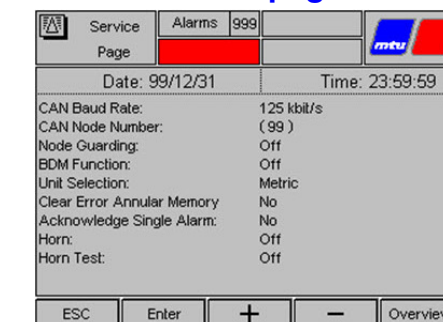
Fuel consumption screen page



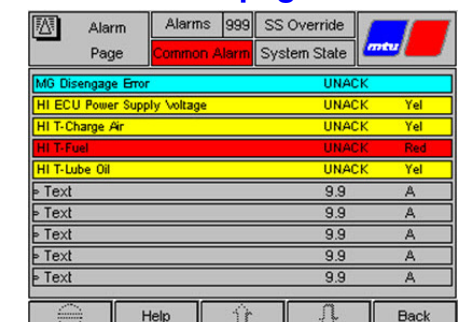
Brightness screen page



Service screen page



Alarm screen page

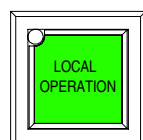




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Local mode

Activation:

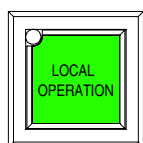


Precondition:
LOCAL
OPERATION
pushbutton is dark.

Press key.

Pushbutton lit.
Local mode activated:
Power train can be controlled by LOP.
Power train can no longer be
controlled via RCS and MCS
(Exception: Emergency stop)

Deactivation:



Press key.

Pushbutton goes out.
Local mode deactivated:
Power train can no longer be
controlled by the LOP.
Power train can be controlled via
RCS and MCS.

Switching power train ready for operation (canceling manual start interlock)



Precondition:
READY FOR
OPERATION
pushbutton dark
(i.e. manual start
interlock is activated).

When the power
train is ready for
operation:
Press pushbutton.

Pushbutton lit.
Start interlock is canceled.

Engine start interlock

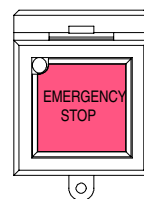


Precondition:
READY FOR
OPERATION
pushbutton lit
(i.e. manual start inter-
lock is not activated).

To prevent engine
start for the power
train:
Press pushbutton.

Pushbutton goes out.
Start interlock is activated.

Emergency engine stop



Precondition:
EMERGENCY STOP
pushbutton is dark.

Lift the protective flap
and press pushbutton
EMERGENCY STOP.

Pushbutton lit.
Alarm signaling at the control
stands (e.g. horn).
Engine is emergency stopped
(by closing emergency air-shutoff
flaps or interrupting fuel supply).

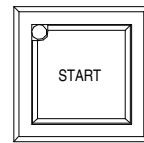
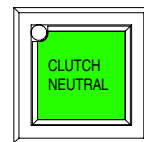
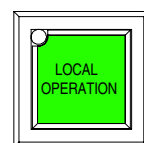
Silence horns at control
stands.
(MCS-6).

Alarm signaling at the control
stands stops.

Alarm acknowledgment
at main control stand
(MCS-6).

“Emergency stop” alarm is
acknowledged.

Engine start



Preconditions:
LOCAL OPERATION
and READY FOR
OPERATION
pushbuttons lit.

With optional PPC:
PPC selector switch set to AUTO.

CLUTCH/NEUTRAL
pushbutton lit.

START pushbutton
dark.

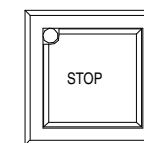
Press pushbutton.

START pushbutton lit.
Automatic starting sequence.
The PPC is switched on as
required (oil priming does not take
place when starting is repeated).
On completion of the starting
sequence:
Engine runs at idle speed.
Pushbutton goes out.
Oil priming is canceled on pressing
the START pushbutton once more
(depending on EIM setting)

Engine shutdown



Precondition:
Engine running.
LOCAL OPERATION
pushbutton lit.



STOP pushbutton
dark.

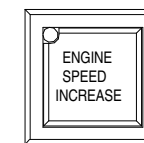
Press pushbutton.

Pushbutton lit. Automatic stopping
sequence. On completion of the
stopping sequence: Engine is
shut down; pushbutton goes out.

Changing engine speed



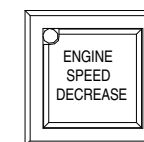
Precondition:
Engine running.
LOCAL OPERATION
pushbutton lit.



ENGINE SPEED
pushbutton dark.

Press and hold down
the pushbutton.

Pushbutton is lit as long as it is
held down.
Engine speed increases.



ENGINE SPEED
pushbutton dark.

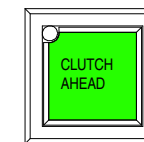
Press and hold down
the pushbutton.

Pushbutton lit as long as it is
held down.
Engine speed decreases.

Changing gearbox state (reversing gear)



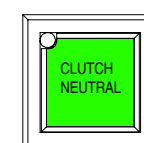
Precondition:
Engine running.
LOCAL OPERATION
pushbutton lit.



CLUTCH AHEAD
pushbutton dark.

Press pushbutton.

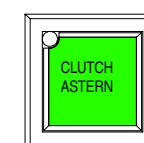
Engine speed in engagement
window. Pushbutton lit.
Gearbox is engaged ahead.



CLUTCH NEUTRAL
pushbutton dark.

Press pushbutton.

Engine speed in engagement
window. Pushbutton lit.
Gearbox is disengaged.



CLUTCH NEUTRAL
pushbutton dark.

Press pushbutton.

Engine speed in engagement
window. Pushbutton lit.
Gearbox is engaged astern.



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Control panel Start/Stop PAN



Light emitting diode

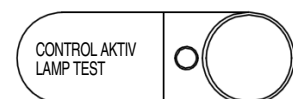
Pushbutton

Transparent sliding cover to prevent inadvertent actuation of the Safety system override and Emergency stop pushbuttons.

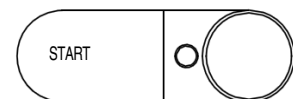
Operating component lamp for displaying the operating status of the PAN

Controls on control panel Start/Stop PAN

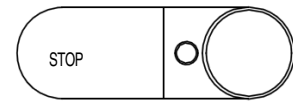
Pushbuttons:



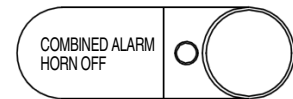
Pressing the pushbutton initiates a lamp test.



Pressing the pushbutton initiates the automatic starting sequence; only possible when CONTROL ACTIVE/LAMP TEST LED is lit green.



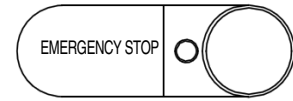
Pressing the pushbutton initiates the stopping sequence.



Pressing the pushbutton switches the horn off.



Pressing the pushbutton overrides the engine safety system.



Pressing the pushbutton initiates an emergency stop.

Displays and control panel Start/Stop PAN



Lit

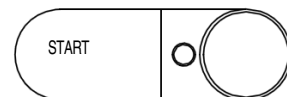


Control stand active. Start possible.

Lit

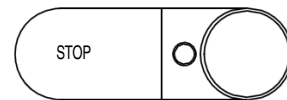


Control stand active. Start not possible, start interlock applied.



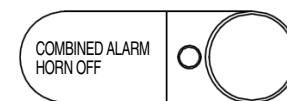
Flashing
Lit

Engine starting sequence running.
Engine running.



Flashing
Lit

Stop procedure running,
When engine speed falls below
300 rpm, or if a stop command is
pending continuously with
the engine stationary.



Flashing
Lit

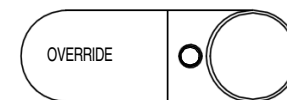


Unacknowledged yellow alarm applied.
Acknowledged yellow alarm applied.

Flashing
Lit



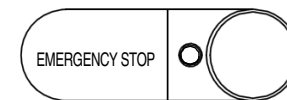
Unacknowledged red alarm applied
Acknowledged red alarm applied.



Lit



Safety system override is active.



Lit



Emergency stop is active.

Operating component lamp Start/Stop PAN

Lit



Normal operation

Flashing (2 Hz)



Boot procedure / communication error

Flashing (8 Hz)



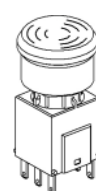
Boot error / serious fault

Off

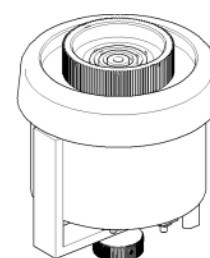


Operating voltage missing / undershot

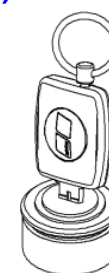
Key switch (Option) and alarm horn (Option)



Alarm horn for
control stands
dia. 28 mm/87dB/IP66



Alarm horn for
control stands
dia. 52 mm/90dB/IP55



- Main key-operated switch
(Master-Key-Switch)
- RCS key-operated switch
(RCS-Key-Switch)

Analog display instruments (optional)

Engine speed



Engine lube oil
pressure



Engine coolant
temperature



Shaft speed



Gear control oil
pressure



Gear oil temperature



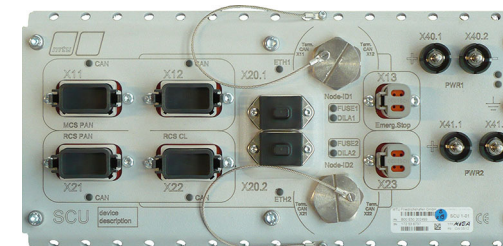
Systembus components in control stand

Ethernet distributor



Switch with following functions:
- Management of Ethernet system bus
- Redundant protection of Ethernet ring
- Service interface

Connection unit in SCU control stand



Systembus Coupling Unit with following functions:
- Interface between system bus and CAB bus
- Connection between control stand components MCS and RCS
- Power supply for control stand components MCS and RCS



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Switching on the overall system, without key switch

Preconditions:
Overall system is switched on with master switch (circuit breaker).

Switch on master switch (circuit breaker) for the power train.

Note:
Remote/Local operating status is stored.
When started, the plant changes automatically to the stored operating status.

- Local Operating Panel LOP of the power train is booted.
- Background illumination of MCS and RCS of the power train lights up.
- Utility lamp of the Start/Stop PAN lights up for the power train concerned.
- Command (CMD) LED on the command units on the control stands of the power train lit (Remote mode). The LED is lit if the plant was in Local mode when last switched off.
- All LEDs on the Start/Stop PAN and the command units lit: Automatic lamp test in progress.

Repeat procedure for other power trains (if applicable). Procedures are repeated in the same way as for the first power train.

Plant with LOP 14:
If the plant was previously not in Local mode when switched off:
• COMMAND LED flashes on those control stands capable of assuming initial command.
• Command unit buzzers sound (providing that RCS Quiet mode is deactivated)

Plant with LOP 13:
• COMMAND LED flashes on those control stands capable of assuming initial command.
• Command unit buzzers sound (providing that RCS Quiet mode is deactivated).

Switching on the overall system, with master/RCS key-operated switch (optional)

Preconditions:
Overall system is switched on with master switch (circuit breaker).



Insert the master/RCS key in the key switch on the main control stand and turn clockwise.

Note:
Remote/Local operating status is stored. When started, the plant changes automatically to the stored operating status.

- Local Operating Panel LOP of the power train is booted
All Local Operating Panels are automatically booted in Standby mode on multiple-engine system (if applicable).
- Background illumination of MCS and RCS of the power train lights up.
- Utility lamp of the Start/Stop PAN lights up for the power train concerned.
- CMD LED on the command units on the control stands of the power train flashes (Remote mode). The LED is lit if the plant was in Local mode when last switched off.
- Buzzers of the command unit sound (provided RCS Quiet Mode is deactivated).

Repeat procedure for other power trains (if applicable).
Procedures are repeated in the same way as for the first power train.

Switching off the overall system, without key switch

Preconditions:
On-board power supply is switched on. The engine comes to a standstill.

Switch off master switch (circuit breaker) for the power train.

- Overall power train system is out of operation.

Repeat procedure for other power trains (if applicable). Procedures are repeated in the same way as for the first power train.

Switching off the overall system, with master/RCS key-operated switch (optional)

Preconditions:
On-board power supply is switched on. The engine comes to a standstill.



Insert the master/RCS key in the key switch on the main control stand and turn counterclockwise.

- Overall power train system is out of operation.
On multi-engine plants, all Local Operating Panels remain in Network-ON (Standby mode), until the last power train has been switched off.

Repeat procedure for other power trains (if applicable). Procedures are repeated in the same way as for the first power train.

Engine startup at the Start/Stop PAN

Preconditions:
• Overall system of the power train is switched on.
• Propulsion control lever on the command unit of the commanding control stand is set to NEUTRAL.
• No engagement request and no start interlocks applied.
• LED CONTROL ACTIVE/LAMP TEST on control panel Start/Stop PAN on the control stand lights up green.

Note: The automatic starting sequence can be canceled at any time by pressing pushbutton STOP.



Press START button.

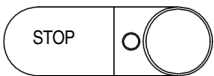
Note:
Starting is not executed if CONTROL ACTIVE/LAMP TEST is yellow when the Start pushbutton is actuated. An appropriate message indicating why starting cannot be executed appears on the display of the control stand.

- LED START flashes while the automatic starting sequence is executed.
- LED START lights up.
- For optional oil priming pump: Engine is primed.
- Engine speed increases.
- Engine runs at idle speed on completion of the starting sequence.
- Power train is ready for engagement.

Repeat procedure for each additional engine (if applicable). Starting sequence is repeated in the same way as for the first engine.

Engine shutdown at the Start/Stop PAN

Preconditions:
• Propulsion control lever of commanding control stand should be set to NEUTRAL.
• LED CONTROL ACTIVE/LAMP TEST on control panel Start/Stop PAN on the control stand lights up green.
• Engine running at idling speed.



Press STOP pushbutton.

- LED STOP flashes while the automatic stopping sequence is carried out.
- Engine speed decreases.
- LED STOP lights up when the engine speed has fallen below 300 rpm.
- The engine is at a standstill when the automatic stopping sequence has been completed and LED STOP goes out.

Repeat procedure for each additional engine (if applicable). The stopping sequence is repeated in the same way as for the first engine.

Emergency engine shutdown at the Start/Stop PAN

Preconditions: Engine running.



Slide the locking device up until it engages.

Press EMERGENCY STOP pushbutton.

- LED EMERGENCY STOP lights up.
- Alarm horn sounds.
- LED COMBINED ALARM / HORN OFF flashes up red.
- On engines with emergency air-shutoff flaps: Flaps close.
- Engine is shut down by stopping fuel injection.

Slide the locking device down until it engages.



Press COMBINED ALARM / HORN OFF pushbutton.

- The alarm horn is silenced.
- LED COMBINED ALARM / HORN OFF flashes up red.
- After acknowledging alarm at display: LED COMBINED ALARM / HORN OFF goes out.

Activating/deactivating System Override at Start/Stop PAN

When Override is active, the system ignores certain operating states which would normally lead to automatic power reduction or engine shutdown:

- Note:
- MTU refuses to accept warranty claims for engine damage incurred while the Override function is active! Activation of the Override function is stored in the Engine Control Unit.
 - With the engine at a standstill: Start interlocks are ignored, e.g. the engine can be started even with the gearbox engaged.
 - With the engine running: Engine operating limit value violations no longer lead to power reduction or engine shutdown.
Exception: Violation of the overspeed limit value still leads to shutdown.

Activating override:

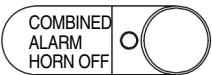


Slide the locking device up until it engages.

Press OVERRIDE pushbutton.

- LED OVERRIDE lights up.
- Alarm horn sounds.
- LED COMBINED ALARM / HORN OFF flashes up red.

Slide the locking device down until it engages.



Press COMBINED ALARM / HORN OFF pushbutton.

- The alarm horn is silenced.
- LED COMBINED ALARM / HORN OFF flashes up red.
- Engine runs in Override mode.
- Rectify any malfunctions and alarm messages manually.
- After acknowledging alarm at display: LED COMBINED ALARM / HORN OFF lights up.

Deactivating override:



Slide the locking device up until it engages.

Press OVERRIDE pushbutton.

- LED OVERRIDE goes out.
- Operating mode Override is deactivated.
- LED COMBINED ALARM / HORN OFF goes out.

Slide the locking device down until it engages.